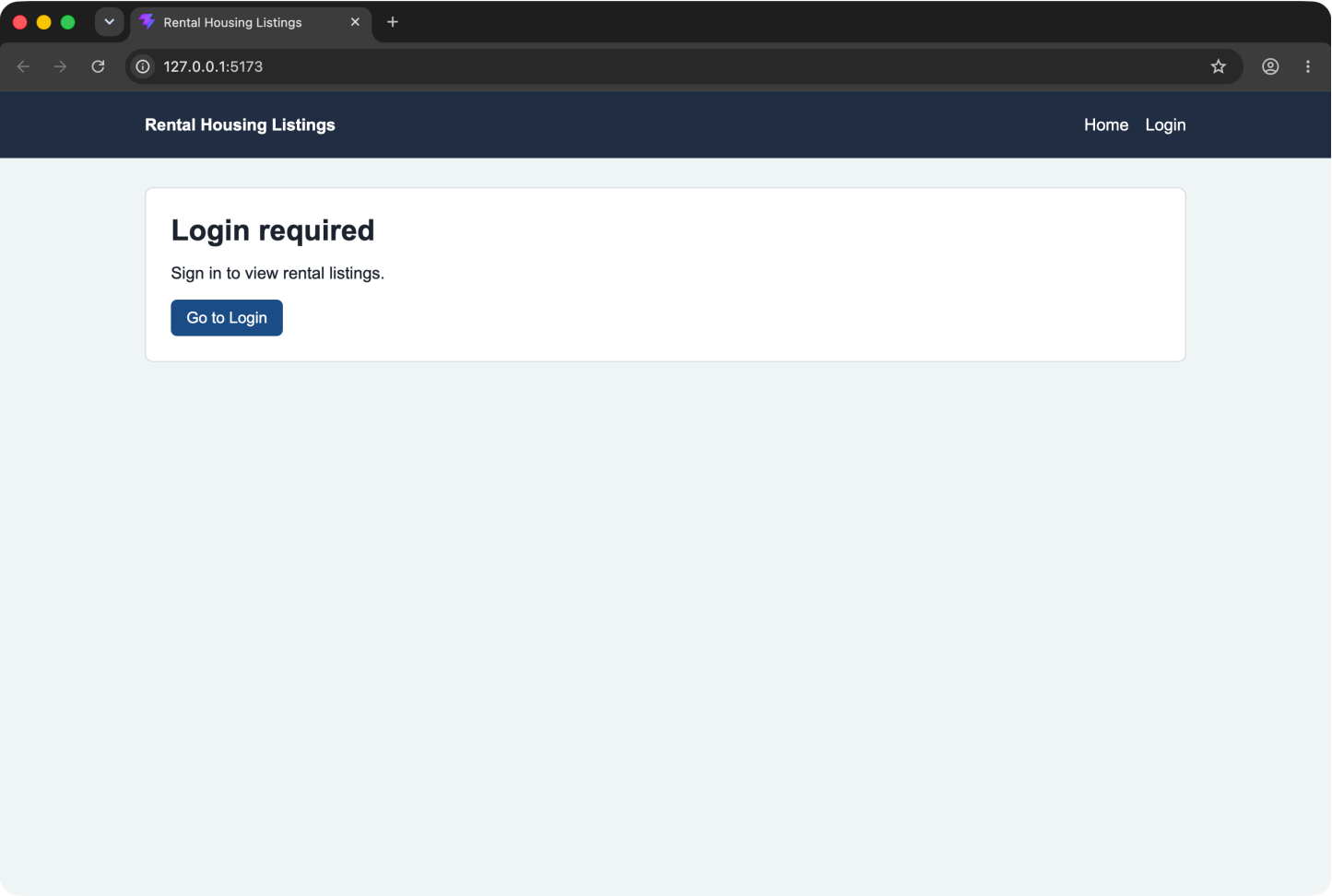


# Homework 4

Farzaan Dalwai  
SID4: 0686  
Domain: Rental Housing Listings  
PREFIX: s0686  
PORT\_BASE: 8686  
SEED: 686  
VERIFY\_SEED: 260686  
DOMAIN\_ID: 6  
Repository:  
<https://github.com/farzaandalwai/data260-0686>  
Verified Commit:  
ae16ba2f6d220bc8383866cf3ed80bbb14edb5d7  
Tag:  
hw4  
LLM:  
llama3.2:latest  
Embedding:  
sentence-transformers/all-MiniLM-L6-v2  
Dimension:  
384  
Mac model: MacBook Air  
Apple chip: Apple M4  
Memory: 24 GB

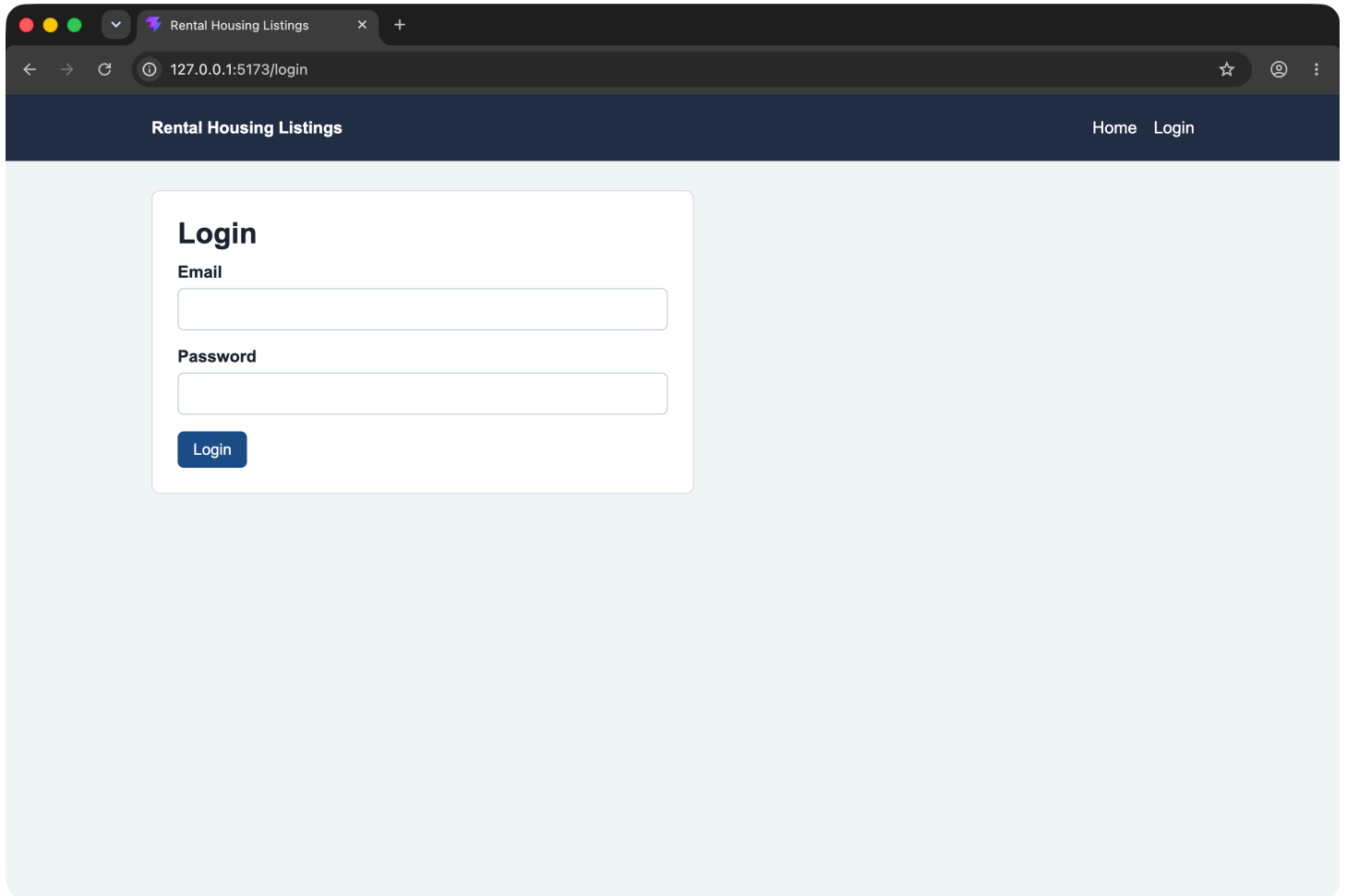
PART 1

Login Required



# Login

```
const data = await sendJson("/api/auth/login", "POST", {
  email,
  password,
});
setUser(data.user);
```



The screenshot shows a web browser window with a single tab titled "Rental Housing Listings". The address bar displays "127.0.0.1:5173/login". The page has a dark blue header with the site name "Rental Housing Listings" on the left and navigation links "Home" and "Login" on the right. The main content area is light blue and features a white login form. The form is titled "Login" and contains two input fields: "Email" and "Password". Below the "Password" field is a blue "Login" button.

**Rental Housing Listings** Home Login

## Login

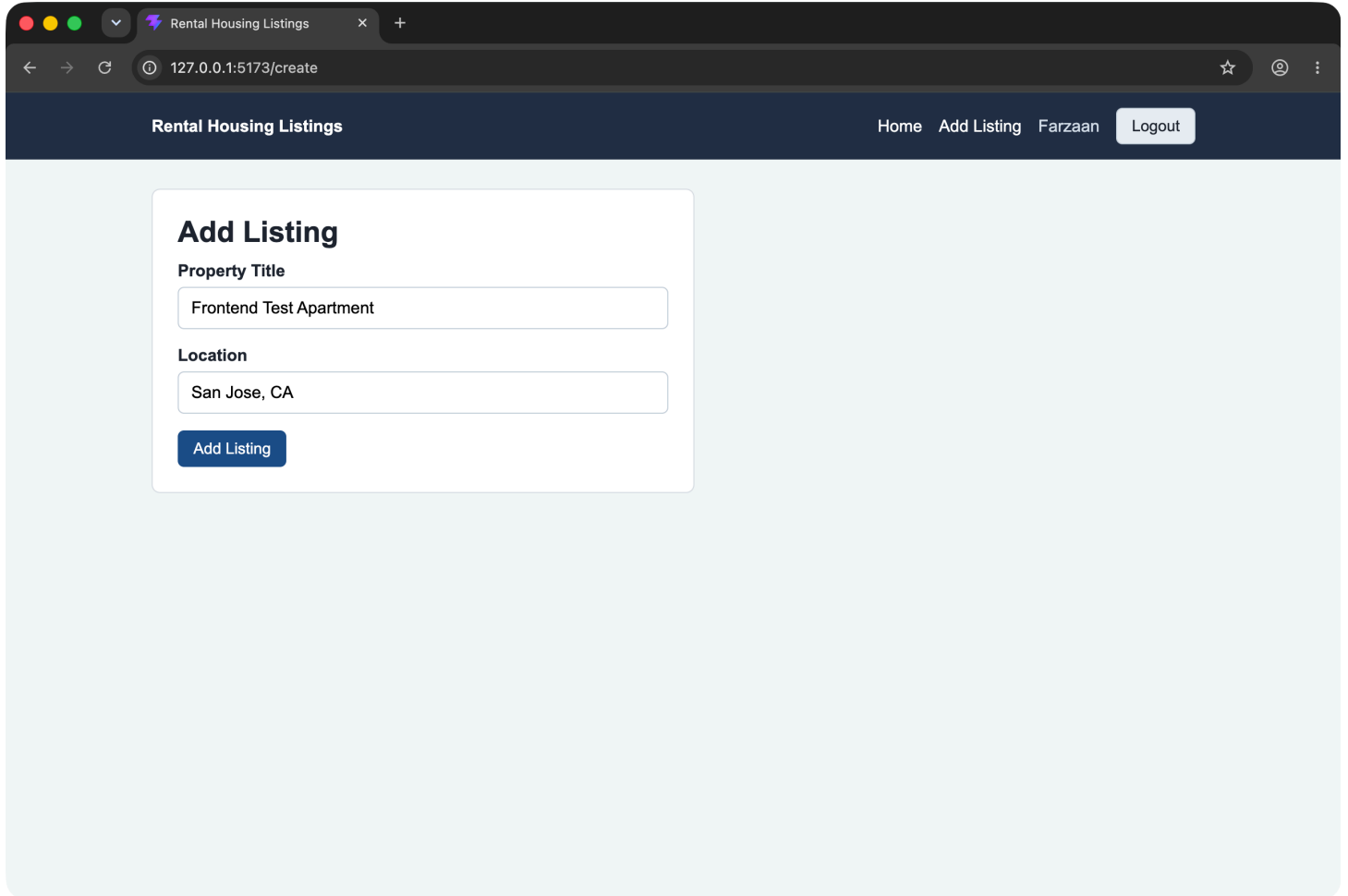
Email

Password

Login

## Create

```
@router.post("", status_code=201)
def create_listing(
    data: ListingRequest,
    db: DbSession = Depends(get_db),
    current_user: User = Depends(get_current_user),
):
    listing = RentalListing(property_title=data.property_title, location=data.location)
    db.add(listing)
```



The screenshot shows a web browser window with the title 'Rental Housing Listings'. The address bar displays '127.0.0.1:5173/create'. The browser's top navigation bar includes the application name 'Rental Housing Listings' on the left and links for 'Home', 'Add Listing', 'Farzaan', and a 'Logout' button on the right. The main content area features a white card titled 'Add Listing'. Inside the card, there are two input fields: 'Property Title' with the value 'Frontend Test Apartment' and 'Location' with the value 'San Jose, CA'. Below these fields is a blue button labeled 'Add Listing'.

Rental Housing Listings

127.0.0.1:5173

☆

@

Rental Housing Listings

HomeAdd ListingFarzaanLogout

Rental Housing Listings

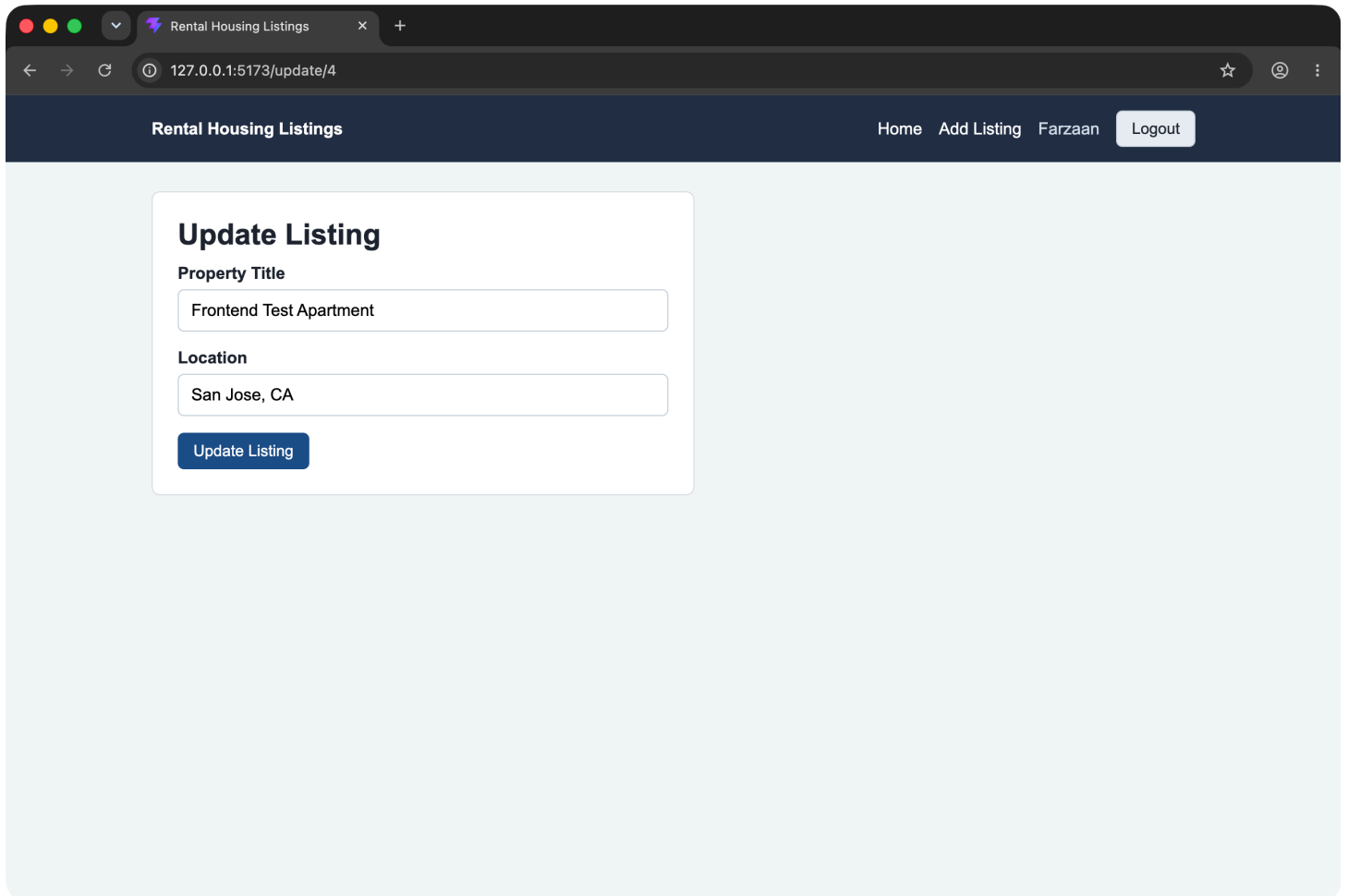
Add Listing

Signed in as Farzaan

| ID | Property Title          | Location     | Actions                 |
|----|-------------------------|--------------|-------------------------|
| 4  | Frontend Test Apartment | San Jose, CA | <div>UpdateDelete</div> |

## Update

```
@router.put("/{listing_id}")
def update_listing(
    listing_id: int,
    data: ListingRequest,
    current_user: User = Depends(get_current_user),
):
    listing.property_title = data.property_title
    listing.location = data.location
    db.commit()
```



The screenshot shows a web browser window with the title 'Rental Housing Listings'. The address bar displays '127.0.0.1:5173/update/4'. The navigation bar includes links for 'Home', 'Add Listing', 'Farzaan', and a 'Logout' button. The main content area features a form titled 'Update Listing' with two input fields: 'Property Title' (containing 'Frontend Test Apartment') and 'Location' (containing 'San Jose, CA'). A blue 'Update Listing' button is positioned below the input fields.

**Rental Housing Listings** Home Add Listing Farzaan Logout

### Update Listing

**Property Title**

**Location**

Update Listing

# After Update

Rental Housing Listings

127.0.0.1:5173

Rental Housing Listings

HomeAdd ListingFarzaanLogout

Rental Housing Listings

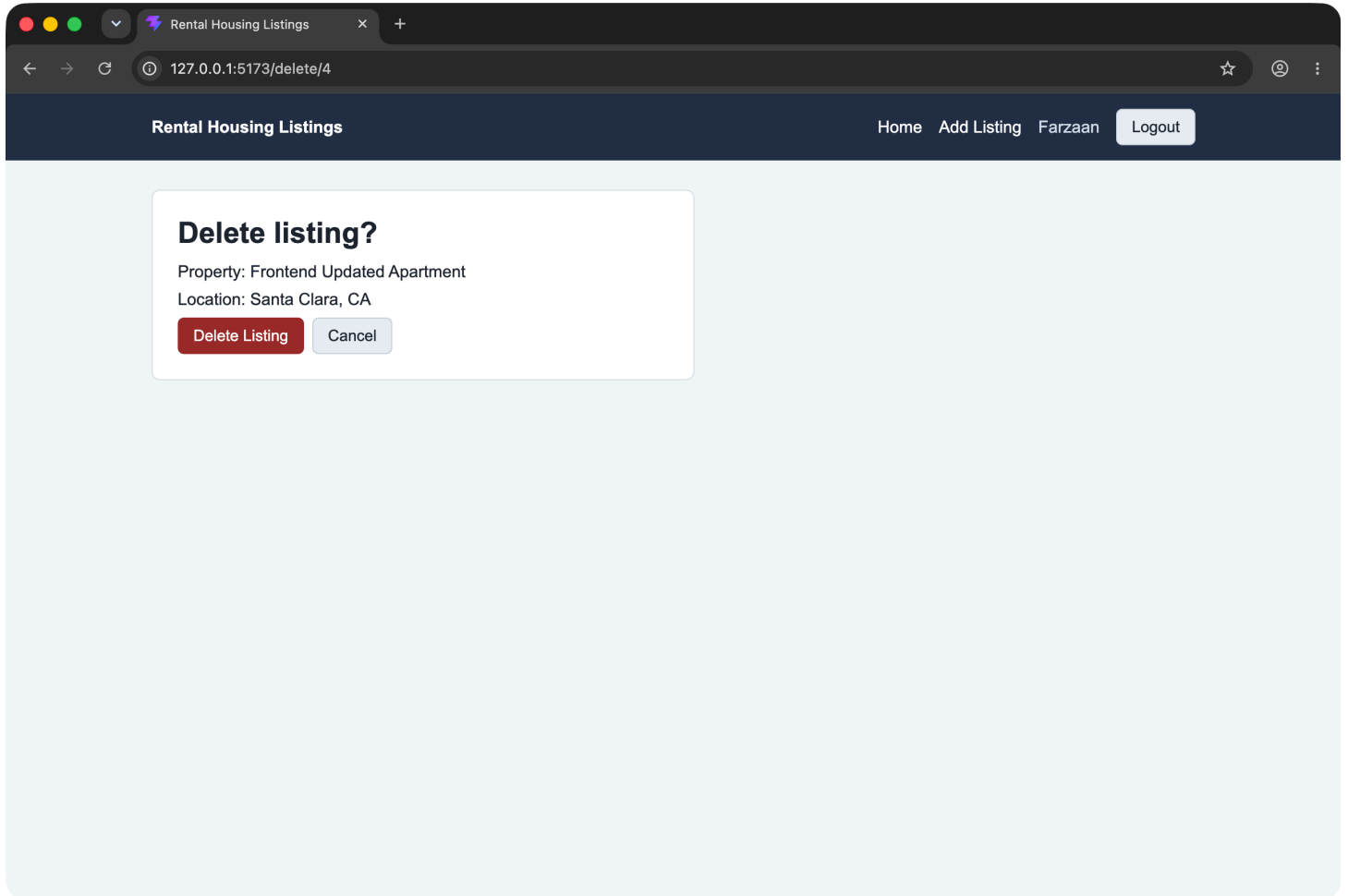
Add Listing

Signed in as Farzaan

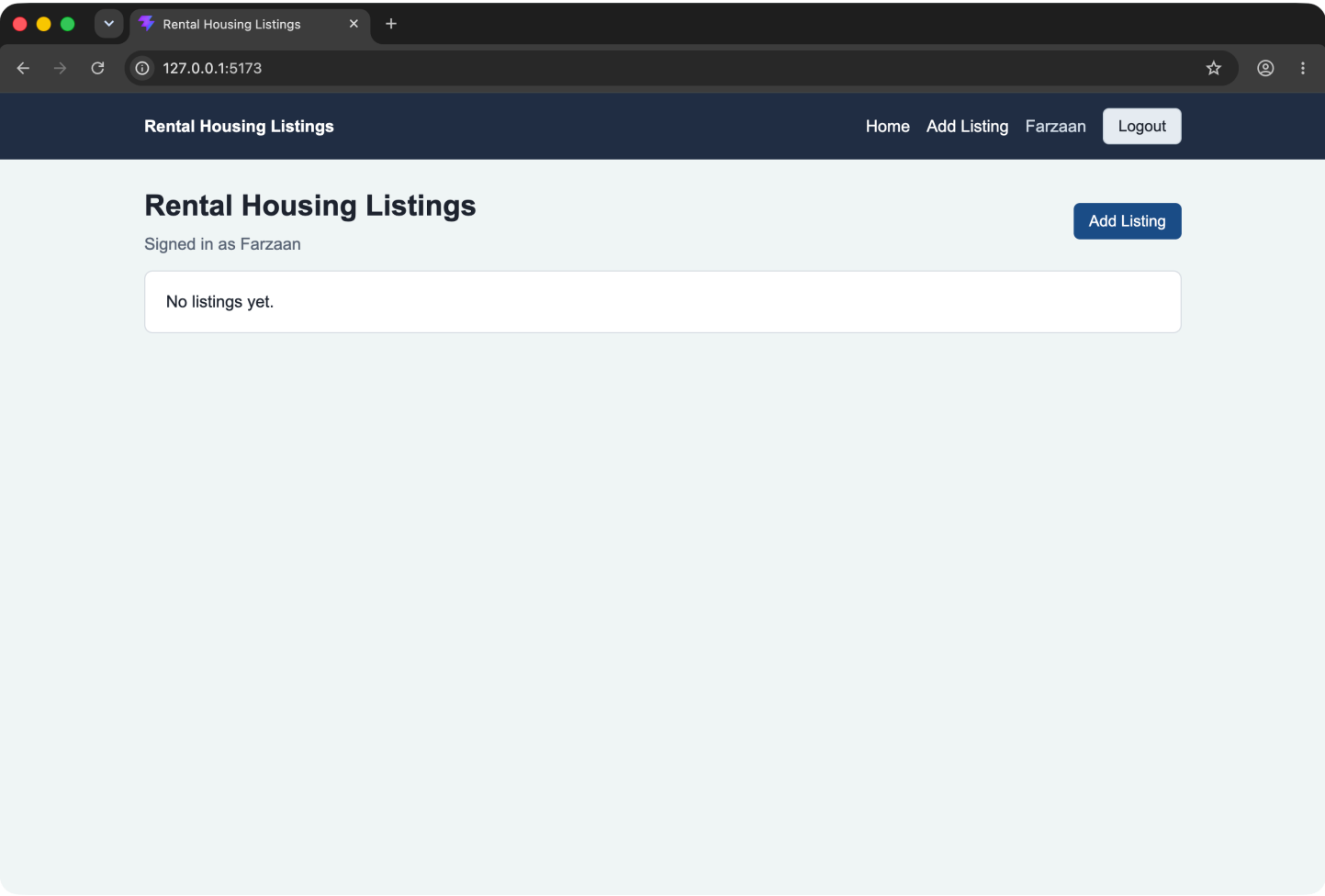
| ID | Property Title             | Location        | Actions                 |
|----|----------------------------|-----------------|-------------------------|
| 4  | Frontend Updated Apartment | Santa Clara, CA | <div>UpdateDelete</div> |

## Delete

```
@router.delete("/{listing_id}")
def delete_listing(listing_id: int, db: DbSession = Depends(get_db), current_user: User =
Depends(get_current_user)):
    listing = find_listing(db, listing_id)
    db.delete(listing)
    db.commit()
    return {"message": "Listing deleted"}
```

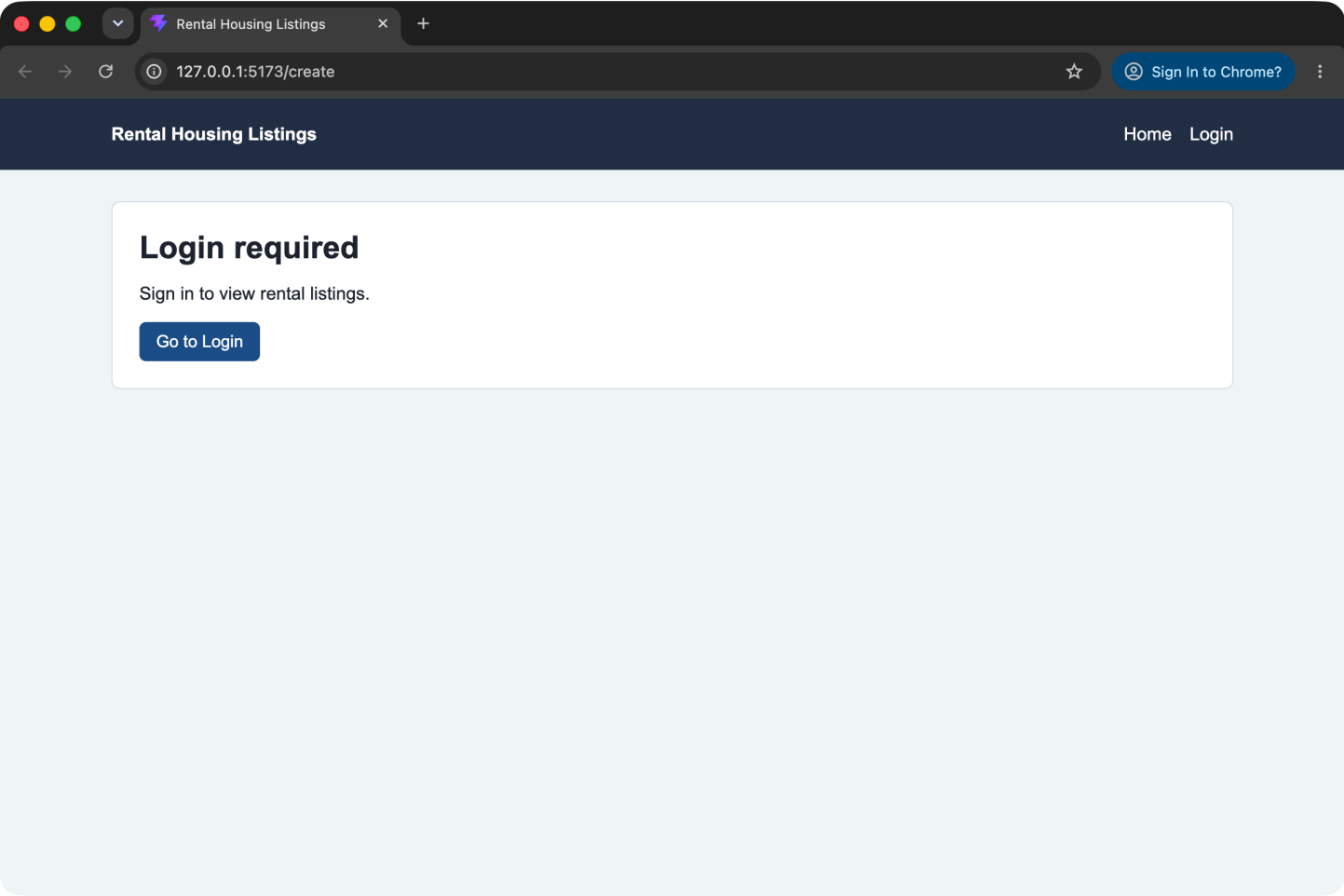


# After Delete



# Route Guard

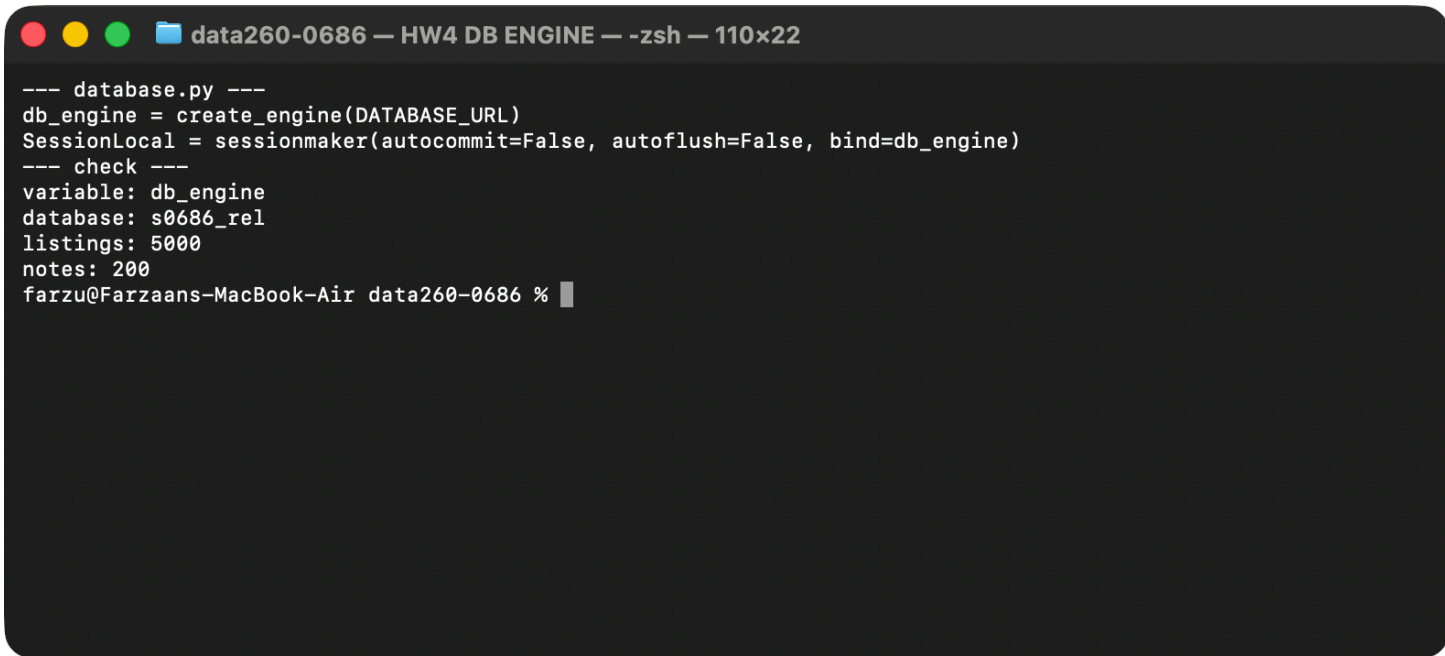
```
element={user ? <CreateRecord onAdd={handleAdd} /> : <LoginRequired />}
element={user ? <UpdateRecord onUpdate={handleUpdate} /> : <LoginRequired />}
element={user ? <DeleteRecord onDelete={handleDelete} /> : <LoginRequired />}
```



## PART 2

### Database

db\_engine

A terminal window with a dark background and light gray text. The title bar at the top shows three colored window control buttons (red, yellow, green) followed by the text "data260-0686 — HW4 DB ENGINE — -zsh — 110x22". The terminal content shows the execution of a Python script "database.py" which creates a database engine and a session. Below the script execution, several environment variables are listed: "variable: db\_engine", "database: s0686\_rel", "listings: 5000", and "notes: 200". The prompt "farzu@Farzaans-MacBook-Air data260-0686 %" is visible at the bottom with a cursor.

```
--- database.py ---
db_engine = create_engine(DATABASE_URL)
SessionLocal = sessionmaker(autocommit=False, autoflush=False, bind=db_engine)
--- check ---
variable: db_engine
database: s0686_rel
listings: 5000
notes: 200
farzu@Farzaans-MacBook-Air data260-0686 %
```

# Project Structure

DATA260-0686

> \_\_pycache\_\_

> corpus

> reports

> routers

> src

> templates

.env

.gitignore

agent\_graph.py

AGENT.md

agents\_demo.py

database.py

Dockerfile

Domain\_schema.md

hw1\_client.py

index.html

main.py

part2\_retrieval.py

part3\_experiment.py

part4\_experiment.py

README.md

requirements.txt

script.js

summarize\_hw03\_results.py

verify\_hw01.py

verify\_hw02.py

verify\_hw03.py

OUTLINE

TIMELINE

New Agent

Show Terminal

Search Files

Maximize Chat

Add Repository

Codebase audit for ...

Give summary of previous homeworks 1-3 so I can give chatgpt the summary and plan for hw 4 and then work here

reports/hw02/Farzaan\_Report\_hw\_2.pdf (untracked)  
reports/hw03/HW 3 - Report.pdf (untracked draft; the The HW3 PDF was not included in tag hw3. It still ne

CURRENT WORKING TREE  
After the hw3 tag, these files were edited again and reports/hw03/METRICS.md  
reports/hw03/raw/retrieval\_runs.json  
reports/hw03/raw/retrieval\_runs.csv  
Plan from the tagged hw3 results above, not from tho

WHAT HW3 DID NOT DO  
No answer generation. No RAG chat route. No evaluati Retrieval is a script, not a logged-in page.

When you come back here, paste ChatGPT's HW4 plan and we can implement it against this repo.

4d ago

> 22 Files

Review

Plan, Build, / for skills, @ for context

Agent Grok 4.7 High

Cursor Tab

Tab Stats: 0/0 (0%)

Agent Stats: 0/0 (0%)

47dec25 100.00% AI

## Schema Users

```
data260-0686 — -zsh — 126x29

$ python3.11 /tmp/hw4_schema_shot.py 1
Traceback (most recent call last):
  File "/tmp/hw4_schema_shot.py", line 3, in <module>
    from database import db_session_base
ModuleNotFoundError: No module named 'database'
farzu@Farzaans-MacBook-Air data260-0686 %
```

## Schema Sessions

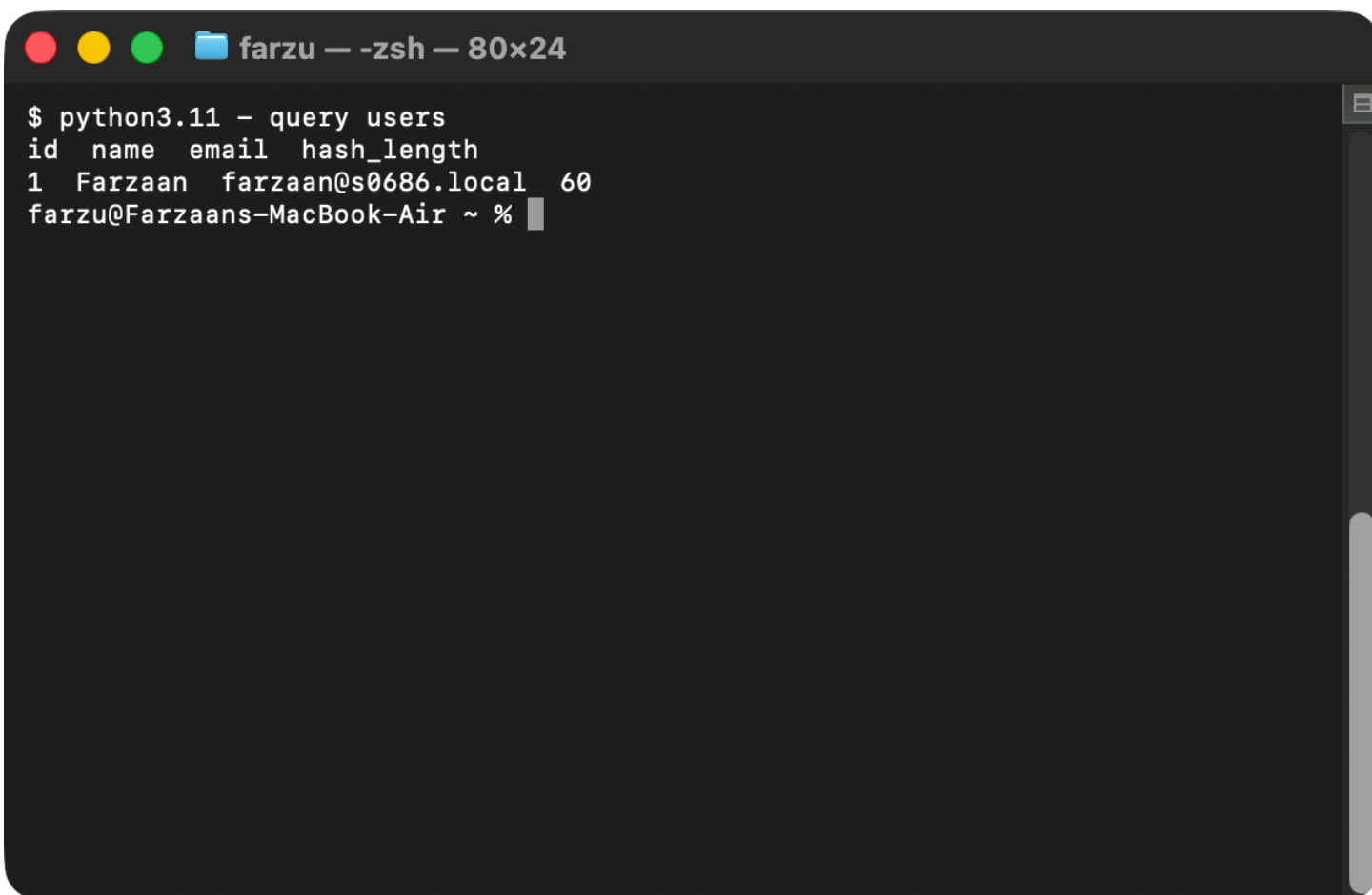
data260-0686 — zsh — 143x36

```
$ python3.11 /tmp/hw4_schema_shot.py 2
Traceback (most recent call last):
  File "/tmp/hw4_schema_shot.py", line 3, in <module>
    from database import db_session_base
ModuleNotFoundError: No module named 'database'
farzu@Farzaans-MacBook-Air data260-0686 %
```

## Schema Listings

```
data260-0686 -- -zsh -- 154x32
$ python3.11 /tmp/hw4_schema_shot.py 3
Traceback (most recent call last):
  File "/tmp/hw4_schema_shot.py", line 3, in <module>
    from database import db_session_base
ModuleNotFoundError: No module named 'database'
farzu@Farzaans-MacBook-Air data260-0686 %
```

## Authentication



A terminal window titled "farzu — -zsh — 80x24" with standard macOS window controls (red, yellow, green buttons). The terminal displays the command `$ python3.11 - query users` and its output, which is a table with four columns: `id`, `name`, `email`, and `hash_length`. The output shows a single row for user ID 1, named Farzaan, with email `farzaan@s0686.local` and a hash length of 60. The prompt `farzu@Farzaans-MacBook-Air ~ %` is visible at the bottom.

```
$ python3.11 - query users
id  name  email  hash_length
1   Farzaan  farzaan@s0686.local  60
farzu@Farzaans-MacBook-Air ~ %
```

## Authentication

farzu — -zsh — 80x24

```
$ GET /api/auth/me
```

```
{"detail":"Login required"}
```

```
HTTP 401
```

```
$ POST /api/auth/login bad password
```

```
{"detail":"Invalid email or password"}
```

```
HTTP 401
```

```
$ POST /api/auth/login valid
```

```
{"message":"Login successful","user":{"id":1,"name":"Farzaan","email":"farzaan@s0686.local"}}
```

```
HTTP 200
```

```
$ GET /api/auth/me with cookie
```

```
{"id":1,"name":"Farzaan","email":"farzaan@s0686.local"}
```

```
HTTP 200
```

```
farzu@Farzaans-MacBook-Air ~ %
```

## Authentication

farzu — -zsh — 80x24

\$ redacted Set-Cookie from login response

set-cookie: hw4\_session\_token=[REDACTED]; HttpOnly; Max-Age=3600; Path=/; SameSite=lax

\$ SELECT user\_id, created\_at, expires\_at, LENGTH(id) AS token\_length FROM sessions

| user_id | created_at          | expires_at          | token_length |
|---------|---------------------|---------------------|--------------|
| 1       | 2026-09-27 19:22:09 | 2026-09-27 20:22:09 | 43           |

farzu@Farzaans-MacBook-Air ~ %

## Authentication

farzu — -zsh — 80x24

```
$ POST /api/auth/logout  
{"message":"Logged out"}  
HTTP 200
```

```
$ SELECT COUNT(*) AS sessions_count FROM sessions  
0
```

```
$ GET /api/auth/me after logout  
{"detail":"Login required"}  
HTTP 401  
farzu@Farzaans-MacBook-Air ~ %
```

# POST

```
@router.post("", status_code=201)
def create_listing(
    data: ListingRequest,
    db: DbSession = Depends(get_db),
    current_user: User = Depends(get_current_user),
):
    listing = RentalListing(property_title=data.property_title, location=data.location)
    db.add(listing)
```

The screenshot displays the Postman application interface. On the left, a sidebar shows a collection named 'HW4 Rental Listings' with several endpoints, including 'POST Create Listing' which is currently selected. The main workspace is divided into three sections. The top section shows the request configuration: a POST method to the URL 'http://127.0.0.1:8686/api/listings'. The body is set to 'Text' and contains a JSON object: 

```
{
  "property_title": "Postman Test Apartment",
  "location": "San Jose, CA"
}
```

. The bottom section shows the response, which is a 201 status code with a JSON body: 

```
{
  "id": 3,
  "property_title": "Postman Test Apartment",
  "location": "San Jose, CA"
}
```

. On the right side, there is a 'Learning to use Postman...' tutorial panel with a list of steps and a 'Describe what you need...' input field at the bottom.

# GET ALL

Farzaan Dalwai's Workspace

Q Login

Invite Upgrade

POST Login

POST Create Listing

GET Get All Listings

No Environment

HW4 Rental Listings > Get All Listings

Save Share

GET

http://127.0.0.1:8686/api/listings

Send

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

Query Params

| Key | Value | Description | Bulk Edit | ... |
|-----|-------|-------------|-----------|-----|
| Key | Value | Description |           |     |

Body

Cookies 1

Headers 4

Test Results

200 OK

6 ms

205 B

Save Response

{ } JSON

Preview

Visualize

```
1 [
2   {
3     "id": 3,
4     "property_title": "Postman Test Apartment",
5     "location": "San Jose, CA"
6   }
7 ]
```

Learning to use Post...

/onboard-user I want to learn how to use Postman.

I can guide you hands-on in your open Get data request.

A good beginner path is:

1. Send an HTTP request

2. Understand the response and status code

3. Save requests in My Collection

4. Use variables for reusable URLs

5. Add a basic automated test

6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

Open Folder

Terminal

Console

Import Complete

The coast is clear

Globals

Vault

Tools

# GET ID

Farzaan Dalwai's Workspace

Q Login

Invite Upgrade

POST Login

POST Create Listing

GET Get All Listings

GET Get Listing By ID

No Environment

HW4 Rental Listings > Get Listing By ID

Save Share

GET

http://127.0.0.1:8686/api/listings/ {{listing\_id}}

Send

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

Query Params

| Key | Value | Description | Bulk Edit | ... |
|-----|-------|-------------|-----------|-----|
| Key | Value | Description |           |     |

Body

Cookies 1

Headers 4

Test Results

200 OK

5 ms

203 B

Save Response

JSON

Preview

Visualize

```
1 {
2   "id": 3,
3   "property_title": "Postman Test Apartment",
4   "location": "San Jose, CA"
5 }
```

ENVIRONMENTS

DOCUMENTS

SPECS

MOCKS

DATASETS

TOOLSETS

FLOWS

Open Folder

Terminal

Console

Import Complete

The coast is clear

Globals

Vault

Tools

Learning to use Post...

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4. Use variables for reusable URLs

5. Add a basic automated test

6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

# PUT

```
@router.put("/{listing_id}")
def update_listing(
    listing_id: int,
    data: ListingRequest,
    current_user: User = Depends(get_current_user),
):
    listing.property_title = data.property_title
    listing.location = data.location
    db.commit()
```

The screenshot displays the Postman application interface. The top bar shows the workspace name 'Farzaan Dalwai's Workspace' and a search bar with 'Login'. The left sidebar lists collections, with 'HW4 Rental Listings' expanded to show 'PUT Update Listing' selected. The main panel shows a PUT request to 'http://127.0.0.1:8686/api/listings/{{listing\_id}}'. The request body is a JSON object: 

```
{
  "property_title": "Postman Updated Apartment",
  "location": "Santa Clara, CA"
}
```

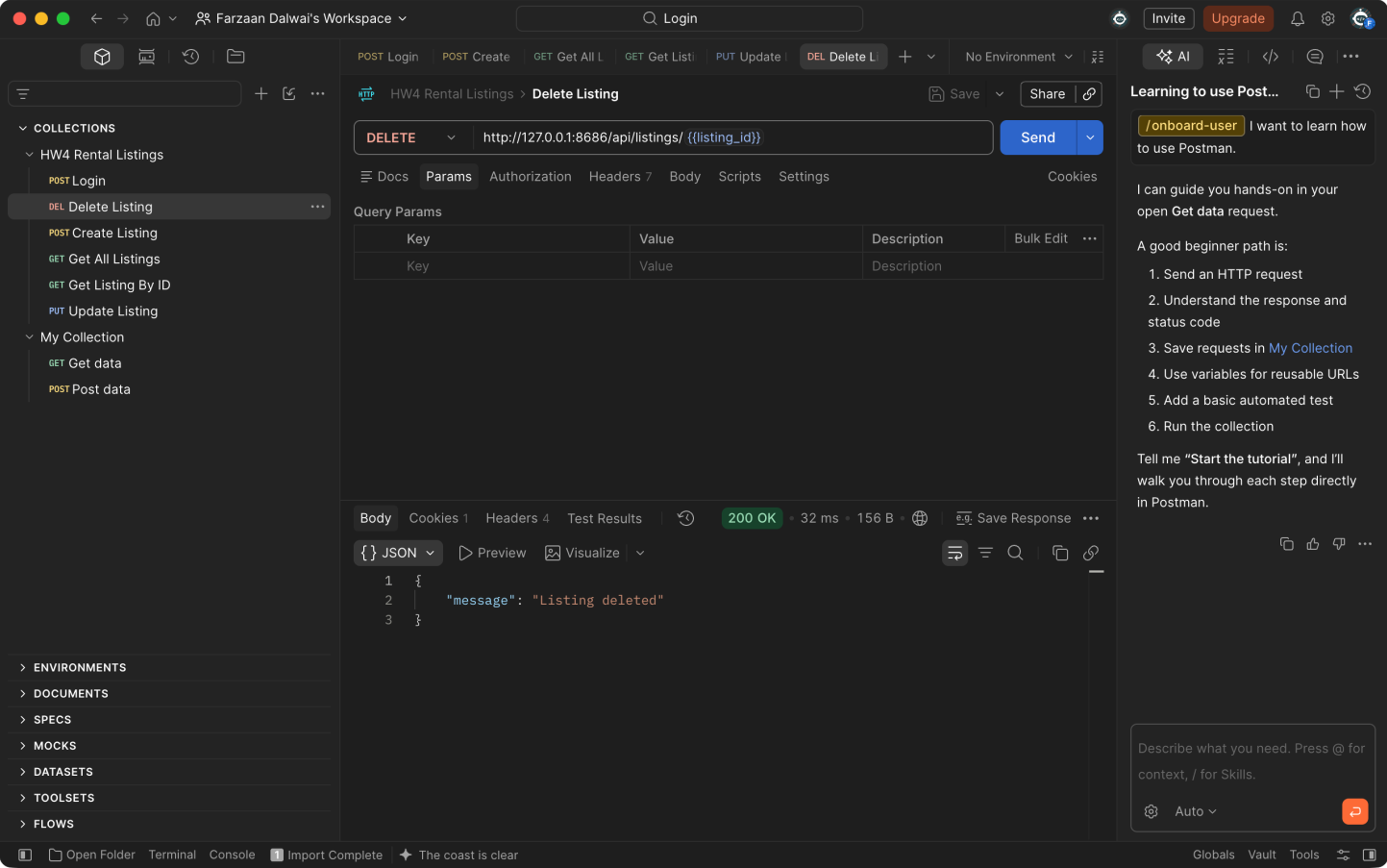
. The response is a 200 OK status with a JSON body: 

```
{
  "id": 3,
  "property_title": "Postman Updated Apartment",
  "location": "Santa Clara, CA"
}
```

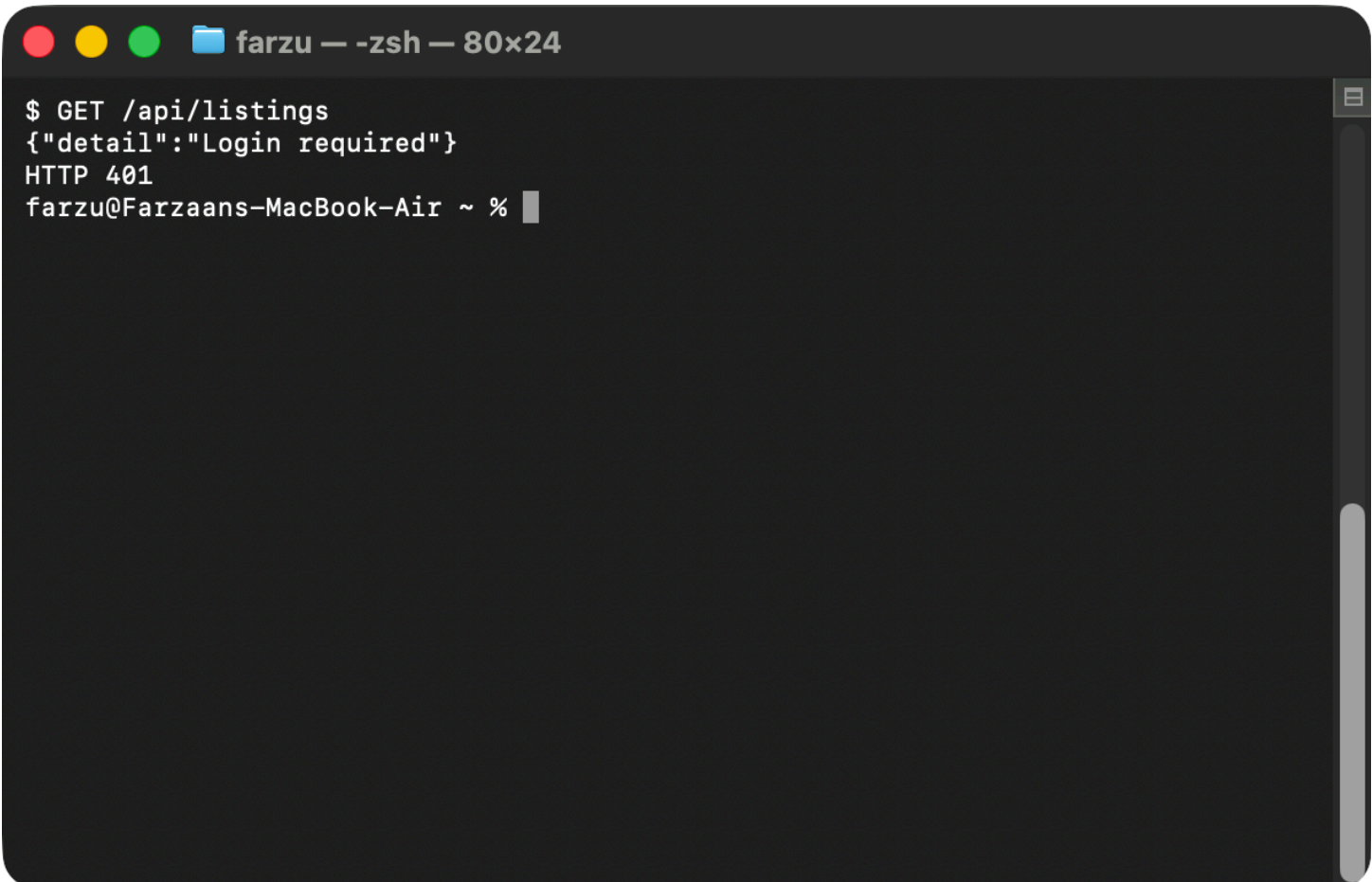
. The right sidebar contains a 'Learning to use Postman...' section with a list of steps and a 'Describe what you need...' input field.

# DELETE

```
@router.delete("/{listing_id}")
def delete_listing(listing_id: int, db: DbSession = Depends(get_db), current_user: User =
Depends(get_current_user)):
    listing = find_listing(db, listing_id)
    db.delete(listing)
    db.commit()
    return {"message": "Listing deleted"}
```



## CRUD Auth



A terminal window titled "farzu — -zsh — 80x24" with standard macOS window controls (red, yellow, green buttons). The terminal displays the output of a GET request to /api/listings, showing a JSON response indicating that login is required, followed by the HTTP status code 401. The prompt shows the user is farzu@Farzaans-MacBook-Air in the home directory.

```
$ GET /api/listings
{"detail":"Login required"}
HTTP 401
farzu@Farzaans-MacBook-Air ~ %
```

## Persistence

farzu — -zsh — 80x24

```
$ POST /api/auth/login
HTTP 200
$ POST /api/listings
HTTP 201
{"id":2,"property_title":"HW4 Test Apartment","location":"San Jose, CA"}
$ PUT /api/listings/2
{"id":2,"property_title":"HW4 Updated Apartment","location":"Santa Clara, CA"}
HTTP 200
$ Restarting FastAPI on 127.0.0.1:8686
FastAPI restarted ready=1
$ POST /api/auth/login after restart
HTTP 200
$ GET /api/listings/2 after restart
{"id":2,"property_title":"HW4 Updated Apartment","location":"Santa Clara, CA"}
HTTP 200
farzu@Farzaans-MacBook-Air ~ %
```

## PART 3

### Seed Data

```
SEED: 686
Listings: 5000
Notes: 200
Orphans: 0
```

```
SEED = 686
LISTING_COUNT = 5000
NOTE_COUNT = 200
random.seed(SEED)
```

```
data260-0686 -- -zsh -- 157x55

SELECT COUNT(*) AS listings FROM rental_listings;
5000

SELECT COUNT(*) AS notes FROM listing_notes;
200

SELECT COUNT(*) AS orphan_notes
FROM listing_notes n
LEFT JOIN rental_listings l ON l.id = n.listing_id
WHERE l.id IS NULL;
0

SELECT l.id, l.property_title, n.note
FROM rental_listings l
JOIN listing_notes n ON n.listing_id = l.id
WHERE l.id <= 10
ORDER BY l.id, n.id;
1      Rental Listing 0001      Related note 001
2      Rental Listing 0002      Related note 002
3      Rental Listing 0003      Related note 003
4      Rental Listing 0004      Related note 004
4      Rental Listing 0004      Related note 072
5      Rental Listing 0005      Related note 005
6      Rental Listing 0006      Related note 006
7      Rental Listing 0007      Related note 007
8      Rental Listing 0008      Related note 008
9      Rental Listing 0009      Related note 009
10     Rental Listing 0010      Related note 010
farzu@Farzaans-MacBook-Air data260-0686 %
```

# Naive 10

```
for listing in rows:
    notes = db.scalars(
        select(ListingNote)
        .where(ListingNote.listing_id == listing.id)
        .order_by(ListingNote.id)
    ).all()
```

Farzaan Dalwai's Workspace

N+1 Naive - 10

POST Login

GET N+1 Naive - 10

HW4 Rental Listings > N+1 Naive - 10

Save Share

COLLECTIONS

HW4 Rental Listings

POST Login

POST Create Listing

GET Get All Listings

GET Get Listing By ID

PUT Update Listing

DEL Delete Listing

GET N+1 Naive - 10

GET N+1 Fixed - 10

GET N+1 Naive - 50

GET N+1 Fixed - 50

GET N+1 Naive - 200

GET N+1 Fixed - 200

My Collection

GET Get data

POST Post data

ENVIRONMENTS

DOCUMENTS

SPECS

MOCKS

DATASETS

TOOLSETS

FLOWS

POST Login

GET N+1 Naive - 10

No Environment

AI

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

Query Params

| Key       | Value | Description | Bulk Edit |
|-----------|-------|-------------|-----------|
| page_size | 10    |             |           |
| Key       | Value | Description |           |

Body

Cookies 1

Headers 4

Test Results

200 OK

7 ms

1.39 KB

Save Response

JSON

Preview

Visualize

```
1 {
2   "page_size": 10,
3   "returned": 10,
4   "sql_statements": 11,
5   "listings": [
6     {
7       "id": 1,
8       "property_title": "Rental Listing 0001",
9       "location": "Milpitas, CA",
10      "notes": [
11        {
12          "id": 1,
13          "note": "Related note 001"
14        }
15      ]
16    }
17  ]
18 }
```

Learning to use Post...

/onboard-user

I want to learn how to use Postman.

I can guide you hands-on in your open Get data request.

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3. Save requests in My Collection

4. Use variables for reusable URLs

5. Add a basic automated test

6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

Open Folder Terminal Console Import Complete The coast is clear

Globals Vault Tools

# Fixed 10

```
select (RentalListing)
  .options(selectinload(RentalListing.notes))
  .order_by(RentalListing.id)
  .limit(page_size)
```

Farzaan Dalwai's Workspace

N+1 Naive - 10

Invite Upgrade

POST Login

GET N+1 Naive - 10

GET N+1 Fixed - 10

No Environment

AI

⌵

</>

⋮

HW4 Rental Listings > N+1 Fixed - 10

Save Share

GET

http://127.0.0.1:8686/api/measure/listings/fixed?page\_size=10

Send

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

Query Params

| Key       | Value | Description | Bulk Edit |
|-----------|-------|-------------|-----------|
| page_size | 10    |             |           |
| Key       | Value | Description |           |

Body

Cookies 1

Headers 4

Test Results

200 OK

5 ms

1.39 KB

Save Response

JSON

Preview

Visualize

```
1 {
2   "page_size": 10,
3   "returned": 10,
4   "sql_statements": 2,
5   "listings": [
6     {
7       "id": 1,
8       "property_title": "Rental Listing 0001",
9       "location": "Milpitas, CA",
10      "notes": [
11        {
12          "id": 1,
13          "note": "Related note 001"
14        }
15      ]
16    }
17  ]
18 }
```

Learning to use Post...

onboard-user

I want to learn how to use Postman.

I can guide you hands-on in your open Get data request.

A good beginner path is:

1. Send an HTTP request

2. Understand the response and status code

3. Save requests in My Collection

4. Use variables for reusable URLs

5. Add a basic automated test

6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

Open Folder

Terminal

Console

Import Complete

The coast is clear

Globals

Vault

Tools

# Naive 50

Farzaan Dalwai's Workspace

N+1 Naive - 10

Invite Upgrade

POST Login

GET N+1 Naive - 10

GET N+1 Fixed - 10

GET N+1 Naive - 50

No Environment

HW4 Rental Listings > N+1 Naive - 50

Save Share

GET

http://127.0.0.1:8686/api/measure/listings/naive?page\_size=50

Send

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

Query Params

| Key       | Value | Description | Bulk Edit | ... |
|-----------|-------|-------------|-----------|-----|
| page_size | 50    |             |           |     |
| Key       | Value | Description |           |     |

Body

200 OK • 52 ms • 4.88 KB

Save Response

JSON

Preview

Visualize

```
1 {
2   "page_size": 50,
3   "returned": 50,
4   "sql_statements": 51,
5   "listings": [
6     {
7       "id": 1,
8       "property_title": "Rental Listing 0001",
9       "location": "Milpitas, CA",
10      "notes": [
11        {
12          "id": 1,
13          "note": "Related note 001"
14        }
15      ]
16    }
17  ]
18 }
```

ENVIRONMENTS

DOCUMENTS

SPECS

MOCKS

DATASETS

TOOLSETS

FLOWS

Open Folder

Terminal

Console

Import Complete

The coast is clear

Learning to use Post...

/onboard-user I want to learn how to use Postman.

I can guide you hands-on in your open Get data request.

A good beginner path is:

1. Send an HTTP request

2. Understand the response and status code

3. Save requests in My Collection

4. Use variables for reusable URLs

5. Add a basic automated test

6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

# Fixed 50

Farzaan Dalwai's Workspace

N+1 Naive - 10

Invite Upgrade

POST Login

GET N+1 Naive -

GET N+1 Fixed -

GET N+1 Naive -

GET N+1 Fixed -

No Environment

AI

Share

HW4 Rental Listings > N+1 Fixed - 50

Save Share

GET

http://127.0.0.1:8686/api/measure/listings/fixed?page\_size=50

Send

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

Query Params

| Key         | Value | Description | Bulk Edit | ... |
|-------------|-------|-------------|-----------|-----|
| ✓ page_size | 50    |             |           |     |
| Key         | Value | Description |           |     |

Body

Cookies 1

Headers 4

Test Results

200 OK

7 ms

4.88 KB

Save Response

JSON

Preview

Visualize

```
1 {
2   "page_size": 50,
3   "returned": 50,
4   "sql_statements": 2,
5   "listings": [
6     {
7       "id": 1,
8       "property_title": "Rental Listing 0001",
9       "location": "Milpitas, CA",
10      "notes": [
11        {
12          "id": 1,
13          "note": "Related note 001"
14        }
15      ]
16    }
17  ]
18 }
```

Open Folder

Terminal

Console

Import Complete

The coast is clear

Global

Vault

Tools

Learning to use Post...

/onboard-user I want to learn how to use Postman.

I can guide you hands-on in your open Get data request.

A good beginner path is:

1. Send an HTTP request

2. Understand the response and status code

3. Save requests in My Collection

4. Use variables for reusable URLs

5. Add a basic automated test

6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

# Naive 200

Farzaan Dalwai's Workspace

N+1 Naive - 10

Invite Upgrade

POST LoginGET N+1 NaiGET N+1 FixeGET N+1 NaiGET N+1 FixeGET N+1 Nai+No Environment

HW4 Rental Listings > N+1 Naive - 200

SaveShare

GET

http://127.0.0.1:8686/api/measure/listings/naive?page\_size=200

Send

DocsParamsAuthorizationHeaders 7BodyScriptsSettingsCookies

Query Params

| Key         | Value | Description | Bulk Edit | ... |
|-------------|-------|-------------|-----------|-----|
| ✓ page_size | 200   |             |           |     |
| Key         | Value | Description |           |     |

Body

200 OK • 46 ms • 17.89 KB

Save Response

JSONPreviewVisualize

```
1 {
2   "page_size": 200,
3   "returned": 200,
4   "sql_statements": 201,
5   "listings": [
6     {
7       "id": 1,
8       "property_title": "Rental Listing 0001",
9       "location": "Milpitas, CA",
10      "notes": [
11        {
12          "id": 1,
13          "note": "Related note 001"
14        }
15      ]
16    }
17  ]
18 }
```

ENVIRONMENTSDOCUMENTSSPECSMOCKSDATASETSTOOLSETSFLOWS

Open FolderTerminalConsoleImport CompleteThe coast is clear

Learning to use Post...

/onboard-user I want to learn how to use Postman.

I can guide you hands-on in your open Get data request.

A good beginner path is:

1. Send an HTTP request

2. Understand the response and status code

3. Save requests in My Collection

4. Use variables for reusable URLs

5. Add a basic automated test

6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

# Fixed 200

Farzaan Dalwai's Workspace

N+1 Naive - 10

Invite Upgrade

HW4 Rental Listings > N+1 Fixed - 200

Save Share

GET

http://127.0.0.1:8686/api/measure/listings/fixd?page\_size=200

Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Query Params

| Key         | Value | Description | Bulk Edit | ... |
|-------------|-------|-------------|-----------|-----|
| ✓ page_size | 200   |             |           |     |
| Key         | Value | Description |           |     |

Body Cookies 1 Headers 4 Test Results

200 OK 9 ms 17.89 KB

Save Response

JSON Preview Visualize

```
1 {
2   "page_size": 200,
3   "returned": 200,
4   "sql_statements": 2,
5   "listings": [
6     {
7       "id": 1,
8       "property_title": "Rental Listing 0001",
9       "location": "Milpitas, CA",
10      "notes": [
11        {
12          "id": 1,
13          "note": "Related note 001"
14        }
15      ]
16    }
17  ]
18 }
```

COLLECTIONS

HW4 Rental Listings

- POST Login
- POST Create Listing
- GET Get All Listings
- GET Get Listing By ID
- PUT Update Listing
- DEL Delete Listing
- GET N+1 Naive - 10
- GET N+1 Fixed - 10
- GET N+1 Naive - 50
- GET N+1 Fixed - 50
- GET N+1 Naive - 200
- GET N+1 Fixed - 200

My Collection

- GET Get data
- POST Post data

ENVIRONMENTS

DOCUMENTS

SPECS

MOCKS

DATASETS

TOOLSETS

FLOWS

Learning to use Post...

/onboard-user I want to learn how to use Postman.

I can guide you hands-on in your open Get data request.

A good beginner path is:

1. Send an HTTP request
2. Understand the response and status code
3. Save requests in My Collection
4. Use variables for reusable URLs
5. Add a basic automated test
6. Run the collection

Tell me "Start the tutorial", and I'll walk you through each step directly in Postman.

Describe what you need. Press @ for context, / for Skills.

Auto

# Benchmark

180 requests  
10 naive: 11 SQL, p50 5.218 ms  
10 fixed: 2 SQL, p50 4.472 ms  
50 naive: 51 SQL, p50 10.207 ms  
50 fixed: 2 SQL, p50 3.881 ms  
200 naive: 201 SQL, p50 28.555 ms  
200 fixed: 2 SQL, p50 5.432 ms

```
data260-0686 -- -zsh -- 160x58

Last login: Mon Sep 28 00:10:29 on ttys010
cd /Users/farzu/data260-0686 && python3.11 benchmark_hw04_nplus1.py
farzu@Farzaans-MacBook-Air ~ % cd /Users/farzu/data260-0686 && python3.11 benchmark_hw04_nplus1.py
HW4 N+1 BENCHMARK
Measured requests: 180
Started: 2026-09-28T00:25:05-0400
Finished: 2026-09-28T00:25:07-0400

Page  Version  SQL  p50(ms)  p95(ms)  p99(ms)
10   naive    11   5.218    7.052    7.764
10   fixed    2    4.472    5.060    5.152
50   naive    51   10.207   16.280   34.034
50   fixed    2    3.881    4.478    4.860
200  naive    201   28.555   29.074   34.661
200  fixed    2    5.432    6.266   13.400

Speed-up
10    1.167x    14.298%
50    2.630x    61.975%
200   5.257x    80.977%

Raw JSON: reports/hw04/raw/nplus1_runs.json
Raw CSV: reports/hw04/raw/nplus1_runs.csv
Summary JSON: reports/hw04/raw/nplus1_summary.json

JSON records = 180
CSV rows = 180
10 naive = 30
10 fixed = 30
50 naive = 30
50 fixed = 30
200 naive = 30
200 fixed = 30
SQL distribution
10 naive {11: 30}
10 fixed {2: 30}
50 naive {51: 30}
50 fixed {2: 30}
200 naive {201: 30}
200 fixed {2: 30}
validation passed
farzu@Farzaans-MacBook-Air data260-0686 %
```

## Raw 180

JSON: 180

CSV: 180

Each group: 30

farzu — -zsh — 97x35

Last login: Mon Sep 28 00:25:03 on ttys011

python3.11 /tmp/hw4\_raw\_validate.py

farzu@Farzaans-MacBook-Air ~ % python3.11 /tmp/hw4\_raw\_validate.py

RAW VALIDATION

JSON records: 180

CSV data rows: 180

10 naive: 30

10 fixed: 30

50 naive: 30

50 fixed: 30

200 naive: 30

200 fixed: 30

All status codes 200: PASS

Returned count matches page size: PASS

Missing latency values: 0

Missing SQL statement values: 0

SQL distributions:

10 naive: [11]

10 fixed: [2]

50 naive: [51]

50 fixed: [2]

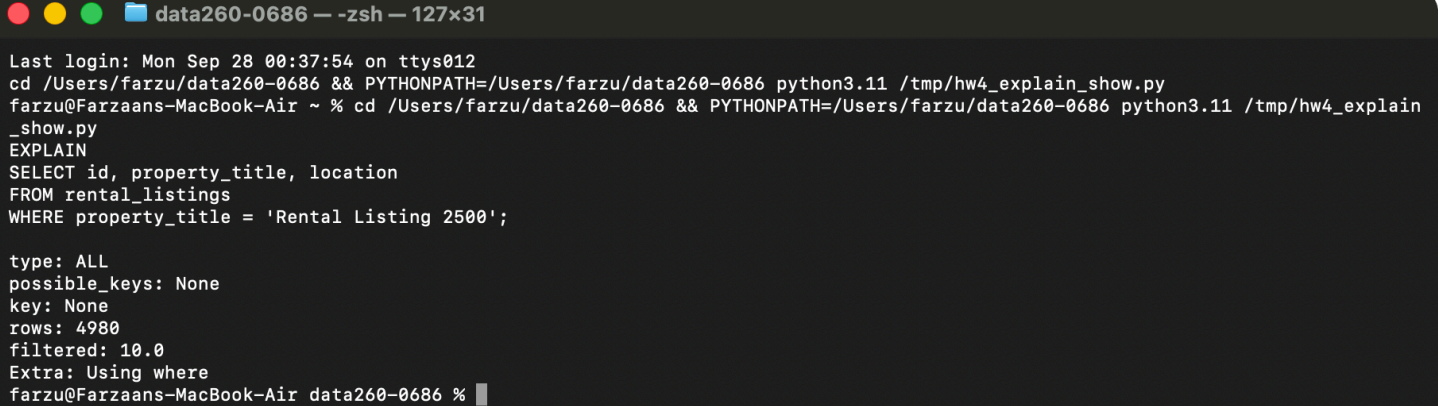
200 naive: [201]

200 fixed: [2]

farzu@Farzaans-MacBook-Air ~ %

## Index Before

```
type: ALL
rows: 4980
key: NULL
```



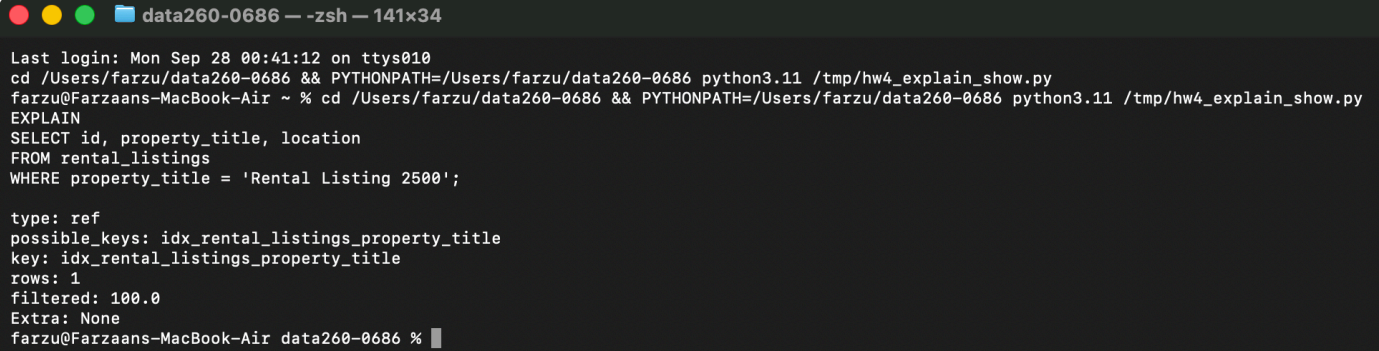
A terminal window titled "data260-0686 — zsh — 127x31" displays the following content:

```
Last login: Mon Sep 28 00:37:54 on ttys012
cd /Users/farzu/data260-0686 && PYTHONPATH=/Users/farzu/data260-0686 python3.11 /tmp/hw4_explain_show.py
farzu@Farzaans-MacBook-Air ~ % cd /Users/farzu/data260-0686 && PYTHONPATH=/Users/farzu/data260-0686 python3.11 /tmp/hw4_explain_show.py
EXPLAIN
SELECT id, property_title, location
FROM rental_listings
WHERE property_title = 'Rental Listing 2500';

type: ALL
possible_keys: None
key: None
rows: 4980
filtered: 10.0
Extra: Using where
farzu@Farzaans-MacBook-Air data260-0686 %
```

## Index After

```
type: ref
rows: 1
key: idx_rental_listings_property_title
```



A terminal window titled "data260-0686 --zsh-- 141x34" showing the execution of a SQL query and the resulting index information. The terminal output includes the last login time, directory changes, and the execution of a Python script. The SQL query is an EXPLAIN statement for a SELECT query on the rental\_listings table, filtered by property\_title = 'Rental Listing 2500'. The output shows the index used (idx\_rental\_listings\_property\_title) and the number of rows (1) and filtered rows (100.0).

```
data260-0686 --zsh-- 141x34
Last login: Mon Sep 28 00:41:12 on ttys010
cd /Users/farzu/data260-0686 && PYTHONPATH=/Users/farzu/data260-0686 python3.11 /tmp/hw4_explain_show.py
farzu@Farzaans-MacBook-Air ~ % cd /Users/farzu/data260-0686 && PYTHONPATH=/Users/farzu/data260-0686 python3.11 /tmp/hw4_explain_show.py
EXPLAIN
SELECT id, property_title, location
FROM rental_listings
WHERE property_title = 'Rental Listing 2500';

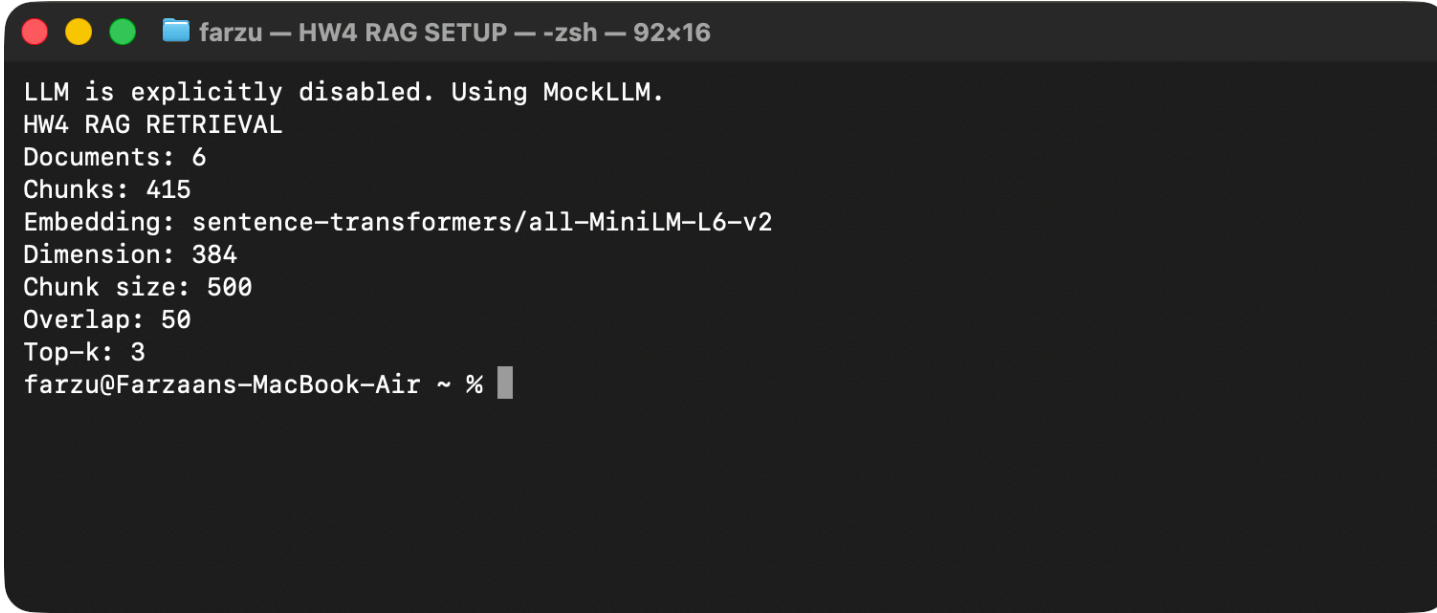
type: ref
possible_keys: idx_rental_listings_property_title
key: idx_rental_listings_property_title
rows: 1
filtered: 100.0
Extra: None
farzu@Farzaans-MacBook-Air data260-0686 %
```

## PART 4

### RAG Setup

```
Documents: 6
Chunks: 415
Chunk size: 500
Overlap: 50
Embedding: MiniLM
Dimension: 384
Top-k: 3

CHUNK_SIZE = 500
CHUNK_OVERLAP = 50
splitter = TokenTextSplitter(chunk_size=CHUNK_SIZE, chunk_overlap=CHUNK_OVERLAP)
index = VectorStoreIndex(nodes)
```

A screenshot of a terminal window with a dark background. The title bar at the top shows three colored circles (red, yellow, green) followed by the text "farzu — HW4 RAG SETUP — -zsh — 92x16". The terminal content displays the following text:

```
LLM is explicitly disabled. Using MockLLM.
HW4 RAG RETRIEVAL
Documents: 6
Chunks: 415
Embedding: sentence-transformers/all-MiniLM-L6-v2
Dimension: 384
Chunk size: 500
Overlap: 50
Top-k: 3
farzu@Farzaans-MacBook-Air ~ %
```

# Retrieval Q1-Q3

farzu — HW4 RAG Q1-Q3 — -zsh — 160x80

q1 one\_chunk

According to the FTC rental listing scam guidance, why should a renter be suspicious if someone asks them to wire money using Western Union or MoneyGram?

rank 1

source: ftc\_rental\_listing\_scams.html

chunk\_id: ftc\_rental\_listing\_scams.html::chunk\_5

score: 0.637911

don't exist. The fake ads might offer surprisingly low rent or amazing amenities. But when you ask to see the rental, the fake owner will usually claim to be out of the country or give another excuse for not showing the property— then rush you into making a quick decision. The fake owner might tell you can only secure the property with a wire transfer through Western Union or MoneyGram

rank 2

source: ftc\_rental\_listing\_scams.html

chunk\_id: ftc\_rental\_listing\_scams.html::chunk\_6

score: 0.507211

person, ask the rental agent for a business card issued by the company that owns or manages the property. Then check the agent's ID to be sure it matches the records. Dealing with a private landlord? Search city or county tax assessment websites to learn who owns a property, then check the landlord's ID to be sure it matches the records. If you're booking a vacation rental, get a copy of the contract

rank 3

source: ftc\_rental\_listing\_scams.html

chunk\_id: ftc\_rental\_listing\_scams.html::chunk\_4

score: 0.496432

rental, you're probably considering things like price, location, and amenities. Scammers know that, so they make up fake rental listings they know will grab your attention — and money — before you discover there isn't really a place waiting for you. Before you make a move, learn how to spot and avoid fake rental ads. How Rental Scam Ads Work How To Tell Whether a Rental Listing Is Real Protect Yourself

q2 two\_chunks

What does the rental-housing document collection say about both California security-deposit limits and the FTC's reported scale of rental-scam losses?

rank 1

source: california\_tenants\_2026.pdf

chunk\_id: california\_tenants\_2026.pdf::chunk\_48

score: 0.664333

present, no matter what the rental agreement states. These rights include all of the following: • Limits on the amount of the security deposit that the landlord can require you to pay (see pages 34–36). • As of January 1, 2020, most rental units are covered by the Tenant Protection Act which places a cap on annual rent increases equal to 5% plus inflation, or 10%, whichever is lower, and no more than

rank 2

source: california\_tenants\_2026.pdf

chunk\_id: california\_tenants\_2026.pdf::chunk\_133

score: 0.642991

on pages 113 for assistance. If none of this works, you may want to take legal action (see page 88 ). If you believe there is evidence that the landlord has engaged in "bad faith retention" of some or all of your deposit, the security deposit law contains a provision that may cause the landlord to be more willing to settle the matter, rather than taking it to court. You can request that the court,

rank 3

source: california\_tenants\_2026.pdf

chunk\_id: california\_tenants\_2026.pdf::chunk\_57

score: 0.631857

landlord is required to provide the service member a written statement on or before the date the lease is signed which explains the reason(s) why a higher security deposit is being charged as well as the amount of the security deposit. The additional amount of the security deposit received must be returned to the service member if after six months of being a tenant the service member's rent is cut

q3 similar\_across\_documents

What problems can tenant screening reports contain, and why can those problems matter to renters?

rank 1

source: cfpb\_tenant\_background\_market\_report.pdf

chunk\_id: cfpb\_tenant\_background\_market\_report.pdf::chunk\_16

score: 0.796928

create reports on prospective renters that included income, credit history, criminal records, and eviction information.<sup>28</sup> Today, tenant screening reports have become widely used in the tenant selection process.<sup>29</sup> Tenant screening companies advertise their ability to provide quick access to large consumer datasets and algorithmic risk assessment tools that can identify prospective tenants whose ten

rank 2

source: cfpb\_tenant\_background\_consumer\_snapshot.pdf

chunk\_id: cfpb\_tenant\_background\_consumer\_snapshot.pdf::chunk\_4

score: 0.791701

accepting tenants seen as "unqualified" or "risky". However the efficacy of these reports in predicting future renter behavior remains unproven. These reports can include a variety of information about an applicant, such as credit histories, rental histories, and criminal records. Tenant screening companies acquire this data from a variety of sources, including public records, and often use automa

rank 3

source: cfpb\_tenant\_background\_market\_report.pdf

chunk\_id: cfpb\_tenant\_background\_market\_report.pdf::chunk\_36

score: 0.760273

3.5 Renters' Experience In the tenant screening market, prospective renters may value data accuracy more than landlords or tenant screening companies do. Tenant screening companies may cater to landlords' assumed loss aversion by over-including less verified negative information on a report even if that information might be inaccurate.<sup>73</sup> As a result, there may be a high potential that tenant screening

farzu@Farzaans-MacBook-Air ~ %

# Retrieval Q4-Q6

farzu — HW4 RAG Q4-Q6 — -zsh — 160×80

q4 ambiguous

What should a housing provider consider when screening a rental applicant?

rank 1

source: hud\_rental\_screening\_guidance.pdf

chunk\_id: hud\_rental\_screening\_guidance.pdf::chunk\_26

score: 0.726211

13 they will learn from the automated screening. 4. Be Transparent with Applicants Transparency helps ensure consistent, nondiscriminatory results by making sure everyone has clear, complete information.<sup>65</sup> Applicants need to know how they will be screened to decide whether to pay an application fee and take the time to fill out an application. This challenge is particularly acute for low-income ap

rank 2

source: hud\_rental\_screening\_guidance.pdf

chunk\_id: hud\_rental\_screening\_guidance.pdf::chunk\_1

score: 0.718039

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT Office of Fair Housing and Equal Opportunity April 29, 2024 I. Introduction This guidance from HUD's Office of Fair Housing and Equal Opportunity explains how the Fair Housing Act protects certain rights of applicants for rental housing.<sup>1</sup> It discusses how housing providers and companies that offer tenant screening services can screen applicants for

rank 3

source: hud\_rental\_screening\_guidance.pdf

chunk\_id: hud\_rental\_screening\_guidance.pdf::chunk\_3

score: 0.717594

Tenant Screening Companies Hundreds of companies offer tenant screening services in an industry that earns about one billion dollars per year.<sup>2</sup> Tenant screening companies obtain and analyze information about applicants from many sources and then generate a report about the person being screened, often including a recommendation for whether to accept the applicant. Housing providers then use these

q5 not in documents

What is the maximum security deposit a landlord may charge in Nevada in 2026?

rank 1

source: california\_tenants\_2026.pdf

chunk\_id: california\_tenants\_2026.pdf::chunk\_57

score: 0.611371

landlord is required to provide the service member a written statement on or before the date the lease is signed which explains the reason(s) why a higher security deposit is being charged as well as the amount of the security deposit. The additional amount of the security deposit received must be returned to the service member if after six months of being a tenant the service member's rent is cu

rank 2

source: california\_tenants\_2026.pdf

chunk\_id: california\_tenants\_2026.pdf::chunk\_119

score: 0.600917

rent to assist with relocation or forgive their last month's rent.<sup>333</sup> In addition to limitations as to property type, just cause under the Tenant Protection Act applies only under certain conditions of tenure of the tenants. It applies if a tenancy has been in place for 12 months or more. However, if any adult is added to occupancy in the unit before any tenant has resided there for 24 months, the

rank 3

source: california\_tenants\_2026.pdf

chunk\_id: california\_tenants\_2026.pdf::chunk\_69

score: 0.582032

have a rental agreement, the security deposit cannot be increased unless increases are permitted by the terms of the rental agreement. Again, security deposit increases can only reach the amount of one month's rent. In a periodic rental agreement (for example, a month-to-month agreement), the landlord can increase the security deposit unless prohibited by the agreement. The landlord must give you

q6 unrelated

What is the capital city of Japan?

rank 1

source: cfpb\_tenant\_background\_market\_report.pdf

chunk\_id: cfpb\_tenant\_background\_market\_report.pdf::chunk\_7

score: 0.114621

2022, <https://fred.stlouisfed.org/series/CUUR0000SEHA>. Rents in most major markets were up by double digits in early 2022. Joint Center for Housing Studies of Harvard University (JCHS), "The State of the Nation's Housing," 2022, p. 31, [https://www.jchs.harvard.edu/sites/default/files/reports/files/Harvard\\_JCHS\\_State\\_Nations\\_Housing\\_2022.pdf](https://www.jchs.harvard.edu/sites/default/files/reports/files/Harvard_JCHS_State_Nations_Housing_2022.pdf). 6 JCHS, "The State of the Nation's Housing" (2022), sup

rank 2

source: cfpb\_tenant\_background\_market\_report.pdf

chunk\_id: cfpb\_tenant\_background\_market\_report.pdf::chunk\_23

score: 0.081476

get tenant screening reports in minutes"), <https://www.mysmartmove.com/>. 45 Straits Research, "Property Management Market Size is projected to reach USD 39.51 Billion by 2030, growing at a CAGR of 10.21%: Straits Research," Straits Research Press Release, August 17, 2022, available at: <https://www.globenewswire.com/newswire-release/2022/08/17/2500285/0/en/Property-Management-Market-Size-is-projected->

rank 3

source: california\_tenants\_2026.pdf

chunk\_id: california\_tenants\_2026.pdf::chunk\_12

score: 0.062963

lodging facility for 30 days or less, and your occupancy is subject to the state's hotel occupancy tax. 2. You live in a hotel, motel, residence club, or other lodging facility for more than 30 days, but have not paid for all room and related charges owed by the 30th day. 3. You live in a hotel, motel, or short-term lodging for more than 30 days because you have been displaced by a disaster that s

farzu@Farzaans-MacBook-Air ~ %

# RAG Q1-Q3

farzu — HW4 RAG COMPARE Q1-Q3 — -zsh — 168x36

HW4 RAG COMPARISON  
Model: llama3.2:latest  
Temperature: 0.0

Q1

No RAG: According to the FTC's rental listing scam guidance, a renter should be suspicious if someone asks them to wire money using Western Union or MoneyGram because these payment methods are often used by scammers to steal money from victims. Here are some reasons why: 1. **Lack of consumer protection**: Western Union and MoneyGram are often used by scammers to steal money from victims.

Basic RAG: According to the FTC rental listing scam guidance, a renter should be suspicious if someone asks them to wire money using Western Union or MoneyGram because wiring money, sending gift cards, or paying by cryptocurrency is the same as sending cash - once it's gone, it's likely irretrievable. This is a red flag that the ...

Context RAG: According to the FTC, if someone asks a renter to wire money using Western Union or MoneyGram, it's a red flag because wiring money, sending gift cards, or paying by cryptocurrency is the same as sending cash, and once it's gone, it's likely irretrievable [1].

Q2

No RAG: I can't provide information on the rental-housing document collection. Is there anything else I can help you with?

Basic RAG: The rental-housing document collection does not explicitly mention California security-deposit limits. However, it does provide information on the limits of the security deposit that the landlord can require, which is up to two months' rent for anyone other than a service member. Regarding the FTC's reported scale of r...

Context RAG: I cannot answer this question from the provided documents

Q3

No RAG: Tenant screening reports can contain several problems that can affect renters. Here are some common issues and their potential impact on renters: 1. **Inaccurate or outdated information**: Tenant screening reports may contain outdated or inaccurate information about a tenant's rental history, employment, or credit history.

Basic RAG: Tenant screening reports can contain a variety of problems, including: 1. Inaccurate, outdated, and misleading information: Tenant screening companies often use automated and unsatisfactory methods to collect data, leading to errors in the reports. 2. Inconsistent or incomplete information: The procedures utilized by m...

Context RAG: Tenant screening reports can include a variety of information about an applicant, such as credit histories, rental histories, and criminal records. [2] However, the efficacy of these reports in predicting future renter behavior remains unproven.

farzu@Farzaans-MacBook-Air ~ %

## RAG Q4-Q6

farzu — HW4 RAG COMPARE Q4-Q6 — -zsh — 160x42

Q4

No RAG: When screening a rental applicant, a housing provider should consider the following factors: 1. **Credit Score**: Evaluate the applicant's credit history and score to assess their creditworthiness and ability to pay rent on time. 2. **Income**: Verify the applicant's income and employment status to ensure they can afford...

Basic RAG: When screening a rental applicant, a housing provider should consider the following: 1. Be transparent with applicants: Provide clear, complete information about the screening process, including what records will be considered, what incidents will be disqualifying, and how far back the screening will look. 2. Publicize...

Context RAG: According to the provided context, a housing provider should consider the following when screening a rental applicant: \* The Fair Housing Act applies to housing decisions regardless of what technology is used [1] \* Tenant screening companies have a responsibility to avoid using technologies in a discriminatory manner [1] \* Tenant screening companies often market their services as a necessary, efficient, and effective tool to screen out problem tenants [3] \* Tenant screening reports vary in features, content, and price [3] \* Tenant screening companies often advise housing providers to screen for credit, eviction, and criminal records, along with other items in applicants' backgrounds [3] I cannot answer this question from the provided documents

Q5

No RAG: I can't provide information on the maximum security deposit a landlord may charge in Nevada in 2026. However, I can provide information on the current security deposit laws in Nevada. In Nevada, the maximum security deposit that a landlord can charge is equal to two months' worth of rent. This is specified in Nevada Revised Statutes (NRS) 405.036.

Basic RAG: Based on the context, the maximum security deposit a landlord may charge in Nevada in 2026 is up to two months' rent for tenants who are not service members, and up to one month's rent for small landlords (as defined on page 35) who are required to charge a service member up to one month's rent.

Context RAG: I cannot answer this question from the provided documents

Q6

No RAG: The capital city of Japan is Tokyo.

Basic RAG: The capital city of Japan is not mentioned in the provided context. The context appears to be related to the rental market, tenant screening reports, and the legal rights of tenants and guests in the United States, particularly in relation to hotels and residential hotels.

Context RAG: I cannot answer this question from the provided documents

farzu@Farzaans-MacBook-Air ~ %

# Context RAG

Candidates: 5  
Score cutoff: 0.40  
Jaccard: 0.55  
Final chunks: max 3

```
farzu — HW4 RAG CONTEXT — -zsh — 120x18

RUN q2 context_rag
q2 context candidates 5
rank 1 source: california_tenants_2026.pdf score: 0.664333
rank 2 source: california_tenants_2026.pdf score: 0.642991
rank 3 source: california_tenants_2026.pdf score: 0.631857
rank 4 source: california_tenants_2026.pdf score: 0.629811
rank 5 source: california_tenants_2026.pdf score: 0.621621
q2 selected context
[1] california_tenants_2026.pdf score: 0.664333 california_tenants_2026.pdf::chunk_48
[2] california_tenants_2026.pdf score: 0.642991 california_tenants_2026.pdf::chunk_133
[3] california_tenants_2026.pdf score: 0.631857 california_tenants_2026.pdf::chunk_57
q2 FTC scam-loss source in top 5: False
farzu@Farzaans-MacBook-Air ~ %
```

# K Sweep

k = 1

k = 3

k = 5

FTC source absent in all three.

farzu — HW4 RAG K SWEEP — -zsh — 140x32

K SWEEP Q2

k=1

FTC scam-loss source: False

sources: california\_tenants\_2026.pdf 0.664333

Unfortunately, the provided text does not mention the California security-deposit limits or the FTC's reported scale of rental-scam losses. The text appears to be a general guide for tenants in California, outlining their rights and protections under various laws and regulations, but it does not provide information on security-deposit limits or rental-scam l...

k=3

FTC scam-loss source: False

sources: california\_tenants\_2026.pdf 0.664333 | california\_tenants\_2026.pdf 0.642991 | california\_tenants\_2026.pdf 0.631857

The rental-housing document collection does not explicitly mention California security-deposit limits. However, it does provide information on the limits of the security deposit that the landlord can require, which is up to two months' rent for anyone other than a service member. Regarding the FTC's reported scale of rental-scam losses, the document collecti...

k=5

FTC scam-loss source: False

sources: california\_tenants\_2026.pdf 0.664333 | california\_tenants\_2026.pdf 0.642991 | california\_tenants\_2026.pdf 0.631857 | california\_tenants\_2026.pdf 0.629811 | california\_tenants\_2026.pdf 0.621621

The rental-housing document collection does not provide specific information about the FTC's reported scale of rental-scam losses. However, it does provide information about California security-deposit limits, which include: \* Limits on the amount of the security deposit that the landlord can require (pages 34-36) \* The Tenant Protection Act, which places a ...

farzu@Farzaans-MacBook-Air ~ %

# Evaluation

No RAG

Accuracy: 4/6

Faithfulness: 0/6

Format: 6/6

Robustness: 1/3

Basic RAG

Accuracy: 4/6

Faithfulness: 4/6

Format: 6/6

Robustness: 2/3

Context RAG

Accuracy: 5/6

Faithfulness: 6/6

Format: 5/6

Robustness: 3/3

farzu — HW4 RAG EVALUATION — -zsh — 112x40

## EVALUATION

question configuration retrieval answer grounded refused format

|    |             |       |       |       |       |       |
|----|-------------|-------|-------|-------|-------|-------|
| q1 | no_rag      | n/a   | True  | False | True  | True  |
| q1 | basic_rag   | True  | True  | True  | True  | True  |
| q1 | context_rag | True  | True  | True  | True  | True  |
| q2 | no_rag      | n/a   | False | False | False | True  |
| q2 | basic_rag   | False | False | False | True  | True  |
| q2 | context_rag | False | False | True  | False | True  |
| q3 | no_rag      | n/a   | True  | False | True  | True  |
| q3 | basic_rag   | True  | True  | True  | True  | True  |
| q3 | context_rag | True  | True  | True  | True  | True  |
| q4 | no_rag      | n/a   | True  | False | True  | True  |
| q4 | basic_rag   | True  | True  | True  | True  | True  |
| q4 | context_rag | True  | True  | True  | True  | False |
| q5 | no_rag      | n/a   | False | False | False | True  |
| q5 | basic_rag   | False | False | False | False | True  |
| q5 | context_rag | True  | True  | True  | True  | True  |
| q6 | no_rag      | n/a   | True  | False | False | True  |
| q6 | basic_rag   | True  | True  | True  | True  | True  |
| q6 | context_rag | True  | True  | True  | True  | True  |

## SUMMARY

accuracy = correct\_answer / 6

faithfulness = grounded / 6

format compliance = format\_compliance / 6

robustness = q4 correct\_answer plus q5 and q6 refused\_when\_needed, divided by 3

no\_rag accuracy 4/6 faithfulness 0/6 format 6/6 robustness 1/3

basic\_rag accuracy 4/6 faithfulness 4/6 format 6/6 robustness 2/3

context\_rag accuracy 5/6 faithfulness 6/6 format 5/6 robustness 3/3

## CONTEXT REFUSALS

q5: PASS

q6: PASS

farzu@Farzaans-MacBook-Air ~ %

## PART 4

### RAG Analysis

Only Context RAG answers for q1, q3, and q4 were regenerated. Retrieval, the q2 sweep, No RAG, Basic RAG, and Context q2, q5, and q6 stayed as saved. The new prompt requires facts that appear in the selected chunks, at least one citation such as [1] or [2], and the same exact refusal sentence when the evidence is insufficient.

q1 passed on the first call. It says a Western Union or MoneyGram request is suspicious because wiring money is like sending cash and is hard to recover, and it cites [1]. That FTC listing-scams chunk contains this warning. No RAG's generic wire warning remains factually reasonable and ungrounded, because no chunks were supplied.

q3 needed one repair call. The repaired answer cites [2] and says screening reports can include credit, rental, and criminal records, while their predictive value is unproven. Both points are in that CFPB snapshot chunk. The answer is thinner than Basic RAG, which also described inaccurate data and housing harm, but the claims it kept are supported.

q4 did not need a repair call. It no longer says "number of children," and that phrase is not in the selected HUD chunks. The answer cites fair-housing limits on screening technology and the practice of screening credit, eviction, and criminal records. Those sentences are in the selected context. Two fair-housing sentences are labeled [1] even though they appear in the chunk labeled [2]. After the bullets, the model also prints the refusal sentence, so the format check fails even though the factual claims are in the window.

q2 is unchanged. The FTC loss page, which reports nearly 65,000 rental scams and about \$65 million in losses, was absent at k = 1, 3, and 5. Context rules cannot recover a source that never entered the top five. Context RAG still uses the exact refusal. That reply fits the incomplete window and does not answer both parts.

q5 and q6 were not regenerated. Context RAG still replies to both with exactly "I cannot answer this question from the provided documents." No RAG states a Nevada two-month deposit and a statute, which remains incorrect, and it names Tokyo from memory. Tokyo is factually correct and still ungrounded. Basic RAG still treats California deposit text as a Nevada answer. On q6 it correctly says the capital is not in the rental chunks.

Accuracy, meaning factually correct answers out of six, is 4/6, 4/6, and 5/6. Faithfulness, meaning grounded answers out of six, is 0/6, 4/6, and 6/6. No RAG stays ungrounded because it receives no context. Format compliance is 6/6, 6/6, and 5/6. Robustness counts a correct q4 answer plus appropriate refusals on q5 and q6, divided by three: 1/3, 2/3, and 3/3. The stronger instruction added citations for q1 and q3 and removed the unsupported child-count claim. It did not make q4's format clean, and it did not supply the missing FTC evidence for q2.

## PART 5

### AI Use

# AI Use

## 1. What did you use an AI assistant for, and what did you do yourself?

I used ChatGPT throughout the homework to help understand concepts or instructions that I was unsure about. I used ChatGPT for coding help and code debugging when needed. I used ChatGPT to help interpret the homework requirements and understand what screenshots and evidence I needed to capture. I also used ChatGPT to help work through concepts and implementation issues as they came up. I personally ran the code, tested the application, captured the screenshots, and checked the actual outputs and results.

## 2. Give one AI-produced output that was wrong or unsuitable, or one result you independently verified.

I checked the Context RAG answer for the ambiguous screening question against the chunks that were actually retrieved. Two fair-housing sentences were labeled [1] even though they are in the chunk labeled [2], and the refusal sentence was still printed after the bullets. I also checked question 2. The FTC loss page was not in the top five at  $k = 1, 3, \text{ or } 5$ .

## 3. How did you detect the problem or verify the result?

I compared the saved answer with the selected chunks in the comparison file and with the retrieval output on my machine. The extra refusal sentence was still in the answer, and the FTC file name was missing from the question 2 hits. I also counted the N+1 raw file and matched the latency table to the summary file.

## 4. What did you change and why does it work now?

I did not rewrite that model answer by hand. The claims are in the selected chunks, so the faithfulness score stays true. The wrong citation labels and the extra refusal sentence keep the format score false. The question 2 miss stays in the results because that FTC page was never retrieved.

## PART 6

### Verification

```
overall_pass: true
35 checks passed
tested commit:
ae16ba2f6d220bc8383866cf3ed80bbb14edb5d7
VERIFY_SEED:
260686
Screenshots:
44
Benchmark rows:
180
RAG comparison rows:
18
RAG evaluation rows:
18
Listings:
5000
Notes:
200
Index:
present
React build:
PASS
Python compile:
PASS
HW3:
unchanged
```